

Predicting Stock Volatility from SEC 10-K Filing Language

CS 129 Project Milestone

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1. Motivation

Public companies disclose risks in their annual 10-K filings, but how informative this language is varies substantially. Some firms use vague boilerplate statements like “general economic conditions may affect results” while others provide firm-specific warnings that give investors a much clearer sense of what exactly the risk is, where it originates from, and how it could quantitatively affect future performance. We investigate whether the linguistic characteristics of these disclosures predict future stock volatility beyond what traditional financial ratios predict.

This is a supervised learning problem with (1) regression to predict realized stock volatility (in the period following the filing), and (2) classification (high vs. low volatility). The practical motivation is that investors could use text-based signals to anticipate risk, and regulators could identify firms with disclosures that lack substantive content. Our core research question is whether text features provide predictive power in addition to purely accounting metrics.

2. Dataset

Data Sources

We collected three data types for S&P 500 companies from 2010-2023:

1. **10-K Filings:** Extracted Item 1A (“Risk Factors”) from SEC EDGAR using the [edgartools](#) Python package. After removing filings with missing or malformed risk sections, we have approximately 4,847 company-year observations.
2. **Stock Returns:** Daily adjusted close prices from Yahoo Finance. We computed realized volatility as the annualized standard deviation of daily returns over the 252 trading days following each 10-K filing date.
3. **Financial Ratios:** Debt-to-equity, return on assets (ROA), current ratio, and log market capitalization, also sourced from Yahoo Finance.

Train/Validation/Test Split

We split by year to preserve real-world ordering of information and prevent temporal leakage:

- **Training:** 2010-2017 (n = 3,412)
- **Validation:** 2018-2019 (n = 847)
- **Test:** 2020-2023 (n = 588, held out until final evaluation)

Exploratory Analysis

Target Distribution: Realized volatility ranges from 0.12 to 1.84 (annualized), with median 0.31 and right skew. We created a binary label where **high_volatility = 1** if volatility exceeds the 75th percentile (0.42).

Text Statistics: Risk factor sections average 8,200 words (std: 3,100). Document length correlates weakly with volatility ($r = 0.08$), suggesting longer disclosures do not predict risk.

Class Balance: 25% of observations are high-volatility. We use class weights in classification models to address this imbalance.

Feature Engineering

We constructed four feature sets:

Feature Set	Dimensions	Description
X_financial	4	Debt/equity, ROA, current ratio, log market cap
X_sentiment	8	Loughran-McDonald positive/negative word %, uncertainty word %, Fog index, word count, sentence count
X_tfidf	500	TF-IDF on unigrams (top 500 by document frequency)
X_finbert	768	Mean-pooled FinBERT embeddings (yiyanghkust/finbert-pretrain)

Sentiment features were implemented from scratch using the Loughran-McDonald financial word lists. FinBERT embeddings were extracted by passing each document through the pre-trained model and averaging token embeddings (we also tested [CLS] pooling with similar results).

3. Method

We applied three model families:

Logistic/Ridge Regression: We tune regularization strength via validation set performance.

Random Forest: We tune `n_estimators` $\in \{100, 200\}$, `max_depth` $\in \{10, 20, \text{None}\}$, and `min_samples_leaf` $\in \{1, 5, 10\}$.

Neural Network (in progress): A 3-layer MLP with batch normalization and dropout, trained with a custom PyTorch training loop.

4. Preliminary Experiments

Key Findings:

- NEED TO ADD

5. Next Steps

Weeks 4-5:

1. Implement neural network with custom training loop: We need to code the forward/backward pass, early stopping, and learning rate scheduling manually in PyTorch instead of using `.fit()`.
2. Hyperparameter sweep for MLP: Grid search over hidden dimensions, dropout rates, and learning rates using validation performance.
3. Final test set evaluation: Run best models on held-out 2020-2023 data to report unbiased performance estimates.
4. Writeup and poster: Finalize figures (ROC curves, feature importance plots, training curves) and complete the report.

6. Contributions

Team Member	Contributions
Juan	Project design, FinBERT feature extraction, baseline model implementation, SEC EDGAR data pipeline
Diya	,Financial ratio collection, exploratory data analysis, Sentiment feature engineering (Loughran-McDonald implementation)
Derek	TF-IDF features, random forest tuning

References

- [1] Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35-65.
- [2] Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.
- [3] Kogan, S., Levin, D., Routledge, B. R., Sagi, J. S., & Smith, N. A. (2009). Predicting risk from 10-K reports. *Proceedings of the 2009 Conference of the North American Chapter of the ACL*, 94-97.